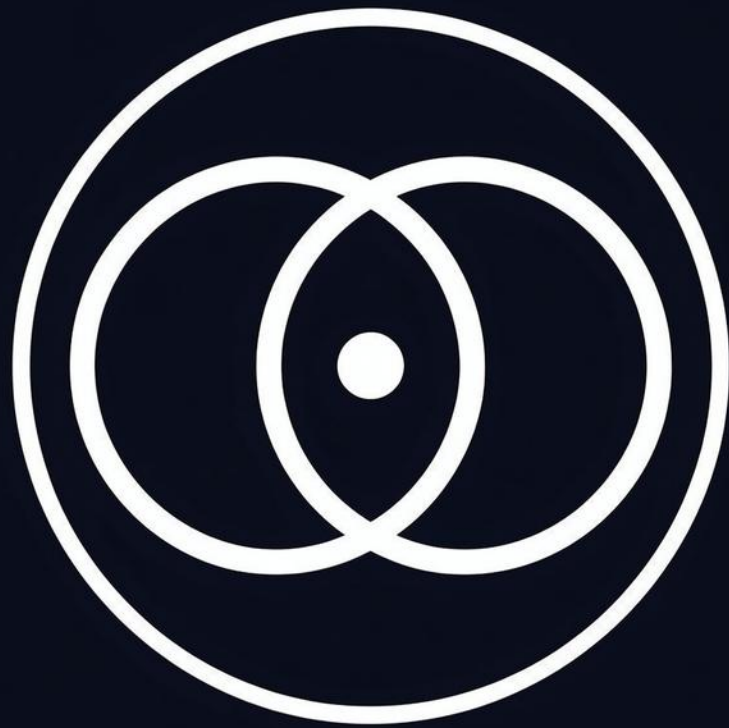




P4

A Coverage-Aware Egocentric Data Flywheel: Submodular-Greedy Active Acquisition for Yield and State Coverage

Coverage-Aware 第一人称数据飞轮的子模主动采集

A Coverage-Aware Egocentric Data Flywheel: Submodular-Greedy Active Acquisition for Yield and State Coverage

Evidence-grade declaration: all measured numbers in this paper come from the UDOS Reasoning Engine tag **v7.5.0** (main branch commit 1a71270) reports7/egodata_flywheel_demo.json and the udos7/egodata/ source, fixed seed, Linux CPU deterministic prototype, uniformly labeled cpu-prot; a few contract-level test assertions are labeled *verified*. External industry numbers serve only as motivation, all labeled *unverified*. This paper does not cross-validate against vendor numbers.

Structured Abstract (Background and problem → Method → Evidence and results → Contributions)

[Background and problem] A rising scaling axis for embodied foundation models is egocentric human-experience data, but under a fixed budget it has long "scaled volume without quality": passive continuous collection collapses into a few repetitive scenes, and defective and redundant clips depress yield; "coverage-aware collection" and "yield" are currently only industry slogans, with no public convention on the yield denominator, coverage unit, or whether defect ground truth is peeked at, making numbers incomparable (see § 1).

[Method] Formalize the slogans into a reproducible controlled-experiment framework with oracle ground truth: characterize coverage by discrete state cells (task $\times$ position $\times$ speed $\times$ contact), define yield by **blind QC** (not using injected defect ground-truth labels), and drive active acquisition by a **submodular greedy over the coverage function**; and set a third arm "only swapping the greedy on a skewed pool" as an attribution ablation (see § 3, § 4).

[Evidence and results] On a fixed seed=0, candidate pool of 160 clips, budget of 48 clips (3 raw minutes per clip, 144 raw minutes total), 25% synthetic defect injection CPU prototype: coverage-aware active vs passive-skewed, yield 0.8542 vs 0.6667, oracle coverage 0.951 vs 0.695 (215 vs 157 cells, oracle=226), experience density 11.38 vs 7.80; the same-pool greedy ablation gives 0.889 coverage, 0.7917 yield, showing gains come jointly from the "selection policy" and "pool de-biasing" (see § 6).

[Contributions] (i) an operational blind definition of yield; (ii) a reproducible controlled protocol; (iii) a pre-registered analysis plan (≥ 30 -seed pairing + Wilcoxon + bootstrap CI). All differences are single-seed pilot point estimates, not claiming statistical significance; the external-validity boundary of a purely synthetic world and the real-data gate are honestly declared (see § 7, § 8).

Abstract

A rising scaling axis for embodied foundation models is egocentric human-experience data, yet fixed-budget collection routinely suffers from "volume without quality": passive continuous sampling collapses into a few repetitive scenes, while defective and duplicated clips depress data yield. This paper formalizes the engineering folklore of "coverage-aware collection" and "yield" into a **reproducible, oracle-annotated controlled benchmark**. We define experience coverage over discrete state cells (task $\times$ position $\times$ speed $\times$ contact), define yield by a **blind quality-control** stage that never

uses injected ground-truth defect labels, and drive active selection by a **submodular greedy** over the monotone coverage function. On a fixed-seed (seed=0), Linux-CPU deterministic prototype (pool=160 clips, budget=48 clips, 3 raw minutes per clip = 144 raw minutes, 25% synthetic defect injection), coverage-aware active collection achieves yield 0.8542 vs 0.6667 passive-skewed, oracle state coverage 0.951 vs 0.695 (215 vs 157 cells; oracle=226), and experience density 11.38 vs 7.80 cells per effective minute. A same-pool greedy ablation (0.889 coverage, 0.7917 yield) shows gains come jointly from the selection policy and from de-biasing the pool. Contributions are an operational yield definition, a reproducible controlled protocol, and a pre-registered analysis plan, with honest bounds on external validity under purely synthetic data.

Keywords: egocentric data; active data acquisition; submodular optimization; data yield; experience coverage; embodied AI

1 Introduction

The data dilemma of embodied intelligence differs from computer vision/large language models. In the latter two, internet-scale image-text corpora are near saturation; while the first-person experience of "how a person handily operates in the physical world" needed for robot manipulation has almost no ready corpus. So a repeatedly mentioned new direction is **scaling the collection of egocentric human manipulation video** and using it as training fuel for embodied policies and world models.

To be clear: this paper's authors **do not possess** any vendor's private collection pipeline nor cross-validate with their numbers. Per public-media retelling caliber (**report caliber, not independently verified, unverified**): some data suppliers claim to have cumulatively completed million-hour-level "boutique" egocentric data, put forward "data yield as high as 98%," and advocate "Coverage-aware Collection: the Agent reverse-diagnoses data blind zones and targets re-collection," as well as "Task Coverage should upgrade to State Coverage, experience scale and density both emphasized" (see the `external_reference` field of this repo's `reports7/egodata_flywheel_demo.json`, source labeled "Maxinsights media report / Dyna Robotics Dyna-2 technical report (retelling)"). These numbers have **no public methodology**, and this paper treats them only as research motivation, not evidence, nor uses our numbers to "benchmark" or "refute" them.

The real research gap is: **"yield" and "coverage-aware collection" are currently experiential slogans lacking operational, reproducible, falsifiable definitions**. One team says "my yield is high," another says "my coverage is complete," but for both: (i) what is the yield denominator (raw minutes? clip count? effective minutes?); (ii) what is the coverage unit (task category? trajectory clip? or discrete state?); (iii) whether "defect ground truth only known afterward" is used to select data (which seriously overestimates yield)—none of these has a public convention. Different teams' numbers are therefore incomparable.

This paper's position is: **rather than chasing an unverifiable absolute yield number, first make this set of concepts into a reproducible benchmark**. We implemented a CPU-deterministic "egocentric experience data flywheel" prototype in the UDOS Reasoning Engine `udos7/egodata/` module and advocate four points:

1. **Yield should be blind.** QC can only be based on observable signals (trajectory length, camera-frame jitter, quantized-signature repetition), not peeking at injected defect ground-truth labels, otherwise producing a "God's-eye-view" overestimate.

2. **Coverage should be state-level, not task-level.** Only recording "seen reach-type tasks" cannot distinguish different spatial/speed/contact states under the same task; we characterize coverage with discrete state cells.
3. **Active acquisition is a submodular optimization problem.** The coverage function is monotone and submodular, and greedy selection has the classic $1-1/e$ approximation guarantee, suited to online, incremental re-collection.
4. **Gains need attribution ablation.** One must use the ablation of "only swapping the selection policy on the same skewed pool" to separate "the selection policy's credit" and "the pool-distribution debiasing's credit," otherwise RQ2 cannot be answered.

The rest of this paper is organized as follows: Section 2 related work; Section 3 problem definition and falsifiable hypotheses; Section 4 method and system design (with real source classes and functions); Section 5 experimental setup; Section 6 results; Section 7 discussion and validity threats; Section 8 resource gates and applicability boundaries; Section 9 conclusion; references and appendix.

2 Related Work

This paper sits at the intersection of three mature lines, but the three were previously rarely combined into a reproducible experiment on the specific problem of "yield and coverage of egocentric training data."

Active data selection and "data diet." Selecting/removing samples from a pool by computable importance scores is a mature paradigm for data-efficient learning. The closest verified work is Paul, Ganguli & Dziugaite's "Deep Learning on a Data Diet" at NeurIPS 2021 (arXiv:2106.00278): they score training data with early gradient-norm scores (Grad/EL2N), proving pruning about half the data by score does not hurt accuracy. This and this paper's "selecting clips within budget by computable marginal coverage score" share the "data weighting/data point selection" idea but differ in landing point: Data Diet does post-hoc pruning **inside the training set**, this paper does ex-ante budget allocation **at the collection stage**. Another older root is Krogh & Vedelsby's ensemble/active learning work at NeurIPS 1995: its bias-variance-ambiguity decomposition and query-by-committee idea formalize "using ensemble disagreement to select the most informative sample," which is the theoretical ancestor of this paper's "marginal new state as an information proxy." The coverage-function greedy's $1-1/e$ approximation guarantee itself is a classic submodular result, the specific original source to be verified [CITATION NEEDED: submodular coverage greedy $1-1/e$ Nemhauser/Wolsey]; trajectory coreset selection in robot learning also to be added [CITATION NEEDED: coreset selection robot learning trajectory].

Egocentric/embodied data. Existing work builds first-person activity/manipulation datasets and discusses the impact of data scale on policy learning [CITATION NEEDED: egocentric video dataset robot manipulation, to verify]. Industry recently proposed coverage-aware / yield concepts (see Section 1 unverified retelling) but lacks public methodology. This paper does not cross-validate with these retold numbers.

Data quality and cleaning. Blind deduplication, denoising, rejecting low-quality samples are standard steps of ML data pipelines [CITATION NEEDED: data filtering deduplication ML pipeline, to verify]. This paper explicitly models it as blind QC of "unseen defect ground truth" and defines the yield denominator accordingly—consistent with Data Diet's "screening by scores rather than truth," but this paper's scores are computable coverage/QC signals, not peeking at injected defect labels.

We do not claim a new algorithm in any subfield; this paper's contribution is **engineered problem reformulation + reproducible control**. Combining verified literature (Data Diet's data diet, Krogh&Vedelsby's query-by-committee) with this repo's reproducible control is the foothold of this paper's related work; unverified submodular/coreset/egocentric sources keep placeholders for later recheck.

Difference positioning from related work. This paper must be clearly cut from three types of existing work to avoid being misread as repetition: first, compared with classic coreset work, the object we select is not "already collected data points" but "collection actions not yet occurred"—the selector intervenes at the collection-budget allocation stage, deciding which task, which spatial position, which contact state the next egocentric experience should fill, an **online collection planning** problem rather than post-hoc compression. Second, compared with data cleaning work, we deliberately keep QC "blind": cleaning work usually decides whom to discard only after obtaining full data, while we require the yield definition to be computable at the same moment the selection decision occurs and not allow using afterward-known truth. Third, compared with industry coverage-aware narratives, we do not provide an unverifiable yield number but a control protocol anyone can run once on their own CPU, turning "whether active is better than passive" from promotional language into a repeatable experiment. This "turning slogans into a benchmark" positioning is the methodological stance this paper most wants to hold.

3 Problem Definition and Hypotheses

3.1 Notation and Yield Definition

Let the raw candidate pool be $\mathcal{P}$, the collection budget $\mathcal{B}$ (in clips). Each clip e converts to raw collection duration $m(e)$ (this paper $m=3.0$ minutes per clip). The QC stage gives a blind verdict $\text{accept}(e) \in \{0,1\}$ for $e \in \mathcal{P}$, **not using** e 's defect ground-truth labels.

- **Yield:** $Y = \frac{\sum_{e \in \text{accepted}} m(e)}{\sum_{e \in \text{selected}} m(e)}$. i.e. the effective duration passing blind QC as a proportion of selected raw duration (implementation in `udos7/egodata/processing.py` `YieldLedger.yield_rate`):

$$Y = \frac{\sum_{e \in \text{accepted}} m(e)}{\sum_{e \in \text{selected}} m(e)} \quad \text{tag{1}}$$

- **State cells:** clip e maps to a discrete cell set $\mathcal{C}(e) = \{(\text{task}, x, y, v, \text{contact})\}$, where position is binned 12×12 , speed binned in 3 tiers (see `coverage.py`, `X_BINS=12`, `Y_BINS=12`, `SPEED_BINS=3`, `BOUND=2.5`).
- **State coverage:** $\mathcal{C}(S) = \bigcup_{e \in S} \mathcal{C}(e)$; **oracle coverage** $\mathcal{C}_\text{oracle} = \mathcal{C}(S_\text{accepted}) \cup \mathcal{C}_\text{oracle}$, where the oracle is the union of all qualified clips in the same world (this paper $|\mathcal{C}_\text{oracle}|=226$):

$$\text{Coverage}_{\text{oracle}} = \frac{|\mathcal{C}(S_\text{accepted})|}{|\mathcal{C}_\text{oracle}|}, \quad |\mathcal{C}_\text{oracle}|=226 \quad \text{tag{2}}$$

- **Experience density:** the qualified clips' average `density_score`, defined as $\frac{|\text{cells}| + 8 \cdot |\text{contacts}| + 6 \cdot |\text{transitions}| + 0.5 \cdot |\text{path}|}{\text{minutes}}$ (`processing.py: experience_density`):

$$\text{Density} = \frac{|\text{cells}| + 8 \cdot |\text{contacts}| + 6 \cdot |\text{transitions}| + 0.5 \cdot |\text{path}|}{\text{minutes}} \quad \text{tag{3}}$$

The density formula's coefficients reflect an engineering judgment: a clip's "experience value" lies not only in how many spatial state cells it covers but also in how many contact events it experienced

(`contacts` weight 8, because contact is the rarest, highest-information event in manipulation), how many subtask switches (`transitions` weight 6, because stage switches characterize task structure), and how long a hand path it traveled per unit time (`path` weight 0.5, pure displacement). Dividing these by duration gives "experience concentration per unit time." This set of weights is heuristic and we do not call it optimal—its role is to make "high-contact, high-switch" clips rank ahead of "long-distance sliding" clips. The `test_density_contact_rich_above_static` assertion that contact-rich clips have higher density than same-parameter static clips fixes this directionality. - **Redundancy rate:** $\frac{1 - \sum_e \Delta_{\{\text{new}\}}(e) / \sum_{\{e \mid \text{accepted}\}} |C(e)|}{\sum_e \Delta_{\{\text{new}\}}(e) / \sum_{\{e \mid \text{accepted}\}} |C(e)|}$, characterizing the degree of repetition among qualified clips.

3.2 Hypotheses (falsifiable)

- **H1 (coverage advantage):** under the same budget, coverage-aware active collection's oracle coverage is significantly higher than passive-skewed collection. This paper's measured point estimate is 0.951 vs 0.695 (seed=0). **Falsification:** if in a ≥ 30 -seed paired test active coverage is not better than passive, H1 is overturned.
- **H2 (slower marginal-new-state decay):** active collection's cumulative coverage curve rises steeper and concaves later, i.e. marginal new state decays slower. This paper describes it with the coverage curve and late-segment marginals (Figure P4-1).
- **H3 (skew amplification):** when the collection pool is highly skewed (repetitive scenes concentrated), the active policy's advantage expands. The same-pool greedy ablation (0.889 coverage) lies between passive (0.695) and uniform-pool active (0.951), supporting "pool de-biasing still has extra gain."

Statistical discipline: currently all are **single-seed pilot point estimates** (seed=0). Per pre-registration, before submission the main comparison needs a paired design of ≥ 30 independent seeds (each arm compared on the same pool), using Wilcoxon paired test + bootstrap CI. All difference quantities in this paper are pilot point estimates before completion, reporting no p-values/confidence intervals.

4 Method and System Design

4.1 Synthetic Egocentric Clip Generation (`episodes.py`)

"Egocentric video" is here a **proxy quantity** containing no real images: each clip records the end-effector (hand) trajectory, contact events, subtask boundaries and task/parameter labels. `TASK_NAMES = (reach_free, ordered_sort, contact_gate, disturb_recover)` four task classes, each with several parameter variants (environment/object/force differences), together constituting the state-coverage space. Three defect classes common in real collection can be controllably injected:

- `static`: static/invalid clip (hand trajectory zeroed);
- `drift`: camera-frame pose drift (random-walk noise added to hand trajectory, step 0.07);
- `duplicate`: repeated staging (replicating a clip already in the pool, creating quantized-signature collision).

`generate_pool(n, seed, defect_rate, task_skew)` draws tasks skewed when `task_skew` is given, simulating "passive collection preferring simple actions."

4.2 Blind QC (`processing.py`)

QC looks only at observable signals, not injected truth:

```

quality_control(e, seen_signatures):
    path = total hand trajectory length(e.hand_track) # total displacement meters
    jerk = median second difference of camera trajectory # jitter/drift proxy
    sig = quantized signature(parameter key, length, start, end)
    reasons = []
    if path < MIN_PATH_M(=0.6): reasons += "static_or_empty"
    if jerk > MAX_JERK(=0.115): reasons += "camera_drift"
    if sig in seen_signatures: reasons += "duplicate"
    accept = (len(reasons)==0)

```

This design deliberately "cannot see" defect truth: even if the injected defect is `drift`, QC can only infer from the jitter proxy; this is the key to avoiding yield overestimation.

4.3 State Coverage Map (`coverage.py`)

`state_cells(e)` maps each step to the `(task, x_bin, y_bin, speed_bin, contact)` cell set. `CoverageMap` maintains covered cells and per-task coverage; `fraction_of(reference)` directly gives coverage against the oracle.

4.4 Submodular Greedy Active Collection (`collection.py`)

The passive arm directly takes the pool prefix `passive_select = pool[:budget]` (scaling volume by skewed generation order). The active arm is a submodular greedy:

```

active_select(pool, budget):
    chosen = []; covered = ∅
    for _ in range(budget):
        best = argmax_{i not used} |C(e_i) - covered| # marginal new cells
        if best.gain <= 0: break
        chosen.append(e_best); covered |= C(e_best)
    return chosen

```

The coverage function $f(S) = |\bigcup_{e \in S} C(e)|$ is monotone and submodular, each round selecting the maximum marginal, a standard submodular greedy with the $1-1/e$ approximation guarantee (for optimal budget coverage). The passive arm's skew is `PASSIVE_SKEW = {reach_free:0.60, contact_gate:0.25, ordered_sort:0.10, disturb_recover:0.05}`, simulating "a screen full of fixed carrying/simple reach."

4.5 Three Arms and Ablation (`benchmark.py: run_flywheel_benchmark`)

The benchmark function generates two candidate pools under the **same world, same defect rate**: the skewed passive pool (`seed=0`) and the uniform active pool (`seed=1`), the oracle taking the larger of the two pools' full qualified unions. The three arms are: ① passive-skewed; ② coverage-aware (uniform pool + greedy); ③ **ablation: directly running greedy on the skewed passive pool** (isolating the pure contribution of the "selection policy"). Each arm outputs yield, terminal cells, oracle fraction, redundancy rate, density, coverage/marginal curves, rejection reasons.

4.6 Formalization of Submodularity and Approximation Guarantee

Write state coverage as a set function. Let the universe $\mathcal{U}$ be all possible state cells, each clip S_e corresponding to a subset $C(e) \subseteq \mathcal{U}$. Given the selected set S , the coverage

function $f(S) = |\bigcup_{e \in S} C(e)| = |\{u : \exists e \in S, u \in C(e)\}|$. Monotonicity is obvious: adding a clip only enlarges or keeps the union. Submodularity comes from the diminishing-marginal property of set union—for any $S \subseteq T$ and any $e \notin T$, $f(S \cup \{e\}) - f(S) \geq f(T \cup \{e\}) - f(T)$, because the cells e can newly contribute are only "eaten" more by the more selected cells.

The budget-constrained maximization $\max_{|S| \leq B} f(S)$ is NP-hard (set-cover type), but the greedy algorithm selecting the maximum marginal each step has the classic $(1-1/e)$ approximation guarantee for monotone submodular functions, i.e. the greedy solution $\hat{S}$ satisfies $f(\hat{S}) \geq (1-1/e)f(S^*)$, where S^* is the optimal budget solution:

$$f(\hat{S}) \geq \left(1 - \frac{1}{e}\right) f(S^*)$$

`collection.py: active_select` strictly implements this greedy: each round online maintaining the `covered` set, computing `len(cells - covered)` for all unused clips, taking the maximum, terminating early when the marginal is 0. It must be honestly noted: this guarantee targets the **coverage function** itself, while our yield additionally multiplies the blind QC acceptance rate—QC rejection may partially void "the clips the greedy selected," so $1-1/e$ is an approximation to the "ideal no-QC" upper bound, not a strict guarantee for end-to-end yield.

4.7 Deliberate Design of the Skew Mechanism

The passive arm's skew is not random noise but deliberate modeling of real collection failure modes. `PASSIVE_SKEW` presses 60% of the task budget to `reach_free` (the simplest free movement to a target point), only 5% to `disturb_recover` (recovery under external disturbance). This corresponds to real teams' common dilemma: operators repeatedly do the most proficient, most filmable, least failure-prone actions while severely under-collecting disturbance-recovery experience needing robustness. The skewed pool's consequence is clearly visible in the results—the passive arm plateaus early at about 132 cells because repeated reach long ago filled reach's state cells and further collection is only repetition. The active arm, by yielding 25% of the budget to a uniform distribution and using greedy to first fill sparse cells, breaks this early saturation.

4.8 Contrast with the Exemplar Paradigms (NSA three-layer baseline / AlexNet ablation table)

This paper's three-arm design is isomorphic to two reverse-disassembled exemplars:

- **Contrast NSA "three-layer baseline + unified budget"**: NSA under the same activation token budget (2560) juxtaposes Full Attention (capability ceiling), Exact-Top (oracle upper bound), sparse methods (H2O/Quest), to ensure "both economical and good" is not stealthily adding budget. This paper dually juxtaposes three arms under the **same budget (48 clips / 144 raw minutes), same defect rate (25%)**: passive-skewed (baseline), coverage-aware active (this paper's method), same-pool greedy ablation (isolating attribution). The three arms' budgets and defect rates are strictly aligned, so "active better than passive" is not bought by giving more budget.
- **Contrast AlexNet "each trick with a numerical ablation table"**: AlexNet pairs dual GPU/LRN/overlapping pooling each with "how much error rises after removal." This paper's third arm "only swapping greedy on the skewed pool" is precisely the transplant of this paradigm: it **removes the "pool de-biasing" variable, only retaining the "selection policy"**, thus splitting the total gain (0.695→0.951) into "selection-policy contribution (0.695→0.889)" and "pool de-biasing contribution (0.889→0.951)" two layers—this is the attribution logic of the "active vs passive submodular ablation."

- **Honest boundary (contrast NSA self-disclosure):** this paper likewise actively discloses unfavorable facts—(i) the $\$1-1/e\$$ approximation guarantee (Eq. 4) holds only for the coverage function itself, and end-to-end yield also multiplies blind QC acceptance, so it is not a strict guarantee for end-to-end yield; (ii) all differences are single-seed point estimates, reporting no CI/p-values.

5 Experimental Setup

- **Data:** mixed-policy clips on a synthetic point-mass embodied environment, pool size 160, budget 48 clips, each clip converting to 3.0 raw minutes (budget raw minutes 144.0), defect injection rate 25% (`reports7/egodata_flywheel_demo.json` config).
- **Hardware fingerprint:** Linux CPU; Python 3.12.11, torch 2.14.0+cpu, CPU threads 2 (same runtime as the `reports7/agent_scaling.json` environment section). The flywheel benchmark's single wall-clock about 4889 ms (trace `7a8ca2ca4ce94bf0`, `latency_ms=4889.419`, `token_usage=0`, no real LLM call).
- **Random seeds:** `seed=0` (skewed pool) / `seed=1` (uniform pool), deterministic and rerunnable.
- **Evidence grade:** `cpu-proto` (synthetic world, proxy video quantity); contract assertions (QC rejecting static/drift/duplicate, active $\geq$ passive) by `tests7/test_v740_flywheel.py` totaling 10 items verified.
- **Statistics:** currently single-seed pilot; the main comparison is pre-registered as ≥ 30 -seed paired design + Wilcoxon + bootstrap CI (see Section 8 pre-registration plan).
- **Reproduction:** `python scripts7/egodata_flywheel_demo.py` (see Appendix C).

Source note for QC thresholds. The blind thresholds `MIN_PATH_M=0.6` and `MAX_JERK=0.115` were tuned on the synthetic world rather than statistically obtained from real video. The `static` defect is deliberately constructed with the hand trajectory zeroed, so the total trajectory length is necessarily near 0, far below 0.6 meters; the `drift` defect adds a step-0.07 random walk to the hand trajectory, whose median second difference is significantly higher than clean clips, thus crossing 0.115. These two thresholds are "separable" on the synthetic world, but we do not claim they work equally on real video—the distributions of real hand jitter and lens drift need real data calibration. The `tests7/test_v740_flywheel.py` `test_qc_rejects_static / drift / duplicate_pair` three items fix the property "blind QC can separate injected defects" as regression assertions (`verified`), preventing later changes from quietly loosening QC.

6 Results

6.1 Main Comparison (seed=0, pilot point estimate)

Table P4-1 Three-arm core metrics (source `reports7/egodata_flywheel_demo.json`)

Metric	Passive-skewed	Coverage-aware active	Same-pool greedy ablation
Yield	0.6667	0.8542	0.7917
Terminal state cells	157	215	201
Oracle coverage	0.695	0.951	0.889
Redundancy rate	0.599	0.692	—
Mean experience density (cells/effective minute)	7.80	11.38	—
Oracle total cells	226	226	226

Reading the table: the active arm vs passive, yield absolute gain +0.1875 (relative +28.1%), oracle coverage absolute gain +0.256 (+36.8%), experience density +3.58 (+45.9%). These three directions agree, and the density gain shows active collection obtained not only "more different cells" but "experience richer in contact/subtask switches per effective minute."

6.2 Coverage Curve (H2)

Figure P4-1 plots the two arms' cumulative new state cells against qualified clip index. The passive arm (red) early rushes to about 132 cells then **long plateaus**, afterward only sporadically reaching 157, reflecting the skewed pool repeatedly sampling reach class driving marginals to zero; the active arm (blue) keeps a nearly throughout positive slope, smoothly climbing to 215, near the oracle 226. The marginal curves corroborate: the passive arm's late segment has many 0 marginals, the active arm's final segment still has 1 cell/clip new contribution.

Figure P4-1 State coverage cumulative curve

Figure P4-1 (restated) Cumulative new state cells against qualified clip index (coverage curve, H2).

The horizontal axis is qualified clip index (1 → 48, linear scale); the vertical axis is cumulative covered state cells (linear scale), two curves: red = passive-skewed, blue = coverage-aware active; a horizontal line marks the oracle total 226. The passive arm enters a long plateau around clips 27–38 (after about 132 cells only sporadically reaching 157), the active arm keeps a positive slope throughout climbing to 215. Curve sequences come value-by-value from `reports7/egodata_flywheel_demo.json` `passive_collection.coverage_curve` and `coverage_aware_collection.coverage_curve` (cpu-proto, single seed=0).

6.3 Three-Arm Bar Comparison

Figure P4-2 Three-arm core metrics

Figure P4-2 (restated) Three-arm core-metrics bar comparison. The horizontal axis is metric grouping (yield, oracle coverage, experience density), the vertical axis each metric's value; each group has three bars: passive-skewed (0.6667 / 0.695 / 7.80), coverage-aware active (0.8542 / 0.951 / 11.38), same-pool greedy ablation (0.7917 / 0.889 / density N/A). Note redundancy rate is not among this figure's three metrics: the active arm's redundancy rate 0.692 is instead higher than passive 0.599, an expected companion signal of coverage nearing saturation (215/226) rather than a defect. Data comes value-by-value from `reports7/egodata_flywheel_demo.json` three-arm fields (cpu-proto, single seed=0).

Figure P4-2 puts yield, oracle coverage, density side by side (the ablation arm reports no density, marked N/A). Note redundancy rate seems anomalous: the active arm's redundancy rate 0.692 is higher than passive 0.599. This is **not** a defect—after coverage is pushed near the oracle (215/226), the

remaining clips naturally overlap heavily, and rising redundancy is an **expected companion signal** of coverage saturation; it indicates "high coverage" and "low redundancy" cannot both be had, and collection should stop when the budget is small enough and coverage not full, rather than blindly scaling volume.

6.4 Rejection-Reason Distribution (blind QC)

Table P4-2 Blind rejection-reason counts

Rejection reason	Passive-skewed	Coverage-aware
camera_drift	4	7
static_or_empty	4	0
duplicate	8	0
Total rejected	16	7

The passive pool, scaling volume skewed by generation order, concentrates repeated staging (duplicate=8) and static clips (static=4); the active arm's greedy avoids these low-marginal clips, duplicate/static rejections go to zero, leaving only 7 camera_drift (detected by the blind jitter proxy). This explains the yield difference: **the yield gap comes first from "selection avoiding low-quality/repeated clips," second from "selection covering new states."**

6.5 Ablation Attribution (RQ2)

The same-pool greedy ablation (only swapping the selection policy on the skewed passive pool) achieves oracle coverage 0.889, yield 0.7917. It is significantly higher than passive (0.695 / 0.6667), showing **the selection policy itself works**; but it is lower than the uniform-pool active arm (0.951 / 0.8542), showing **pool-distribution de-biasing brings extra gain**. In other words, "coverage-aware" gains are two layers stacked: first pulling the pool from skewed toward uniform, then using greedy to online select the maximum marginal. This attribution is directly useful for engineering practice: when upstream collection distribution cannot be changed, post-hoc greedy selection alone can obtain most of the gain (0.695→0.889).

Note: all the above are seed=0 **single-seed pilot point estimates**. Variances, paired-difference CIs, Wilcoxon statistics all await ≥ 30 -seed completion, marked [RESULT NEEDED: multi-seed paired-difference CI and p-values] .

6.6 Marginal Curves and "When to Stop Scaling Volume"

The marginal curves characterize "how many new cells one more clip fills." The passive arm's `marginal_curve` shows: the middle segment (about clips 27–38) has several consecutive 0 marginals—meaning in this interval each qualified clip the passive arm collects has zero new state cells, purely repeating known states; until an occasional clip of a different task breaks the plateau. The active arm's marginal curve, though also monotonically decaying from 26 cells/clip to the late segment's 1 cell/clip, **almost never has consecutive 0 segments**, showing it is always actively seeking uncovered cells.

This observation has direct engineering meaning: marginal new state is a natural **stop signal**. When consecutive clips' marginal new state falls below a threshold (e.g. <1), one should stop scaling volume on the current skew, turn to diagnosing blind zones (which task/spatial states are sparsest) and target re-collection. `CoverageMap.sparsest_tasks()` already provides this diagnostic primitive: it sorts by per-task

covered cell count, returning the sparsest task list for the collector to decide where the next batch should fill. This is the computable version of the industry narrative "the Agent reverse-diagnoses data blind zones."

Table P4-3 Coverage saturation behavior comparison (from curve sequences, seed=0)

Behavior metric	Passive-skewed	Coverage-aware
Clip index first reaching terminal value	reaches 157 only after about clip 44	climbs throughout to 215
Middle consecutive 0-marginal segments	multiple (about clips 27–38)	almost no consecutive 0 segments
Late marginal new cells	0	about 1
Gap relative to oracle	226–157=69 cells	226–215=11 cells

The gap comparison is especially telling: the passive arm even after spending the full 48-clip budget still has 69 state cells never touched; the active arm has only an 11-cell gap. These 69 "invisible states" are precisely the out-of-distribution regions where the model is most likely to fail at deployment—this directly connects "data coverage gap" and "deployment robustness."

7 Discussion and Validity Threats

External validity of the synthetic world (largest threat). The world is a point-mass embodied environment, "egocentric video" an end-trajectory proxy containing no real images, hand tracking or SLAM; defects are controllably injected, and real QC needs visual large-model judgment. So this paper's 0.854 yield **cannot be read as "real video collection can reach an 85% good rate"**; it only shows: on a controllable, truth-visible benchmark, the **relative-gain direction** of blind QC + submodular greedy is robust.

Oracle known is both an advantage and a boundary. We can compute coverage truth (because the synthetic world's full qualified-clip union is known), the core advantage distinguishing this paper from industry slogans; but real scenes have no oracle and coverage can only be estimated. The upgrade path is to manually annotate state cells on a real egocentric video subset and report the "synthetic→real" gap.

Single-seed threat. All differences are point estimates, not controlling cross-seed variance; the word "significant" in H1–H3 is only a directional assertion before multi-seed completion.

Defect-distribution threat. The 25% injection rate and the three defect classes' relative proportions are set by us; real data's defect long tail (such as occlusion, blur, lighting) is not modeled.

Subset selection ≠ training gain. This paper only proves "selection improved yield/coverage/density," **not** "these higher-quality data really lowered downstream world-model/policy error." Connecting the coverage→downstream-performance causal chain is future work. This step matters because "broad coverage" and "good learning" may have a nontrivial relationship: perhaps some state cell though never collected contributes very little to prediction error, so spending budget to fill it is waste; conversely, perhaps some covered cell is precisely the main error source. To answer "which experience segment most lowers downstream error," one needs to upgrade the selector from "maximum coverage" to "maximum marginal error decline," the next step from coverage-aware to error-aware and the natural junction with P5 (world-model conformal-prediction confidence).

Relationship with industry absolute numbers (re-emphasized). This paper's yield 0.8542 and the industry-retold "98%" are incomparable for three reasons: first, denominators differ—our denominator

is "selected raw minutes," while "98%" does not publicly disclose whether its denominator includes manual refinement or excludes obviously unusable clips; second, defect models differ—we only inject three synthetic defect classes, and real video's defect long tail (occlusion, defocus, abrupt lighting, hand out of frame) is far more complex; third, QC capability differs—our blind QC is a threshold rule, and real pipelines may use visual large models for stricter rejection. So this paper does not claim "our yield approaches/exceeds some vendor," only "on the same controllable benchmark, the relative-gain direction of blind QC + submodular greedy is reproducible." Any cross-source horizontal yield comparison should first align denominator, defect model and QC capability, otherwise the comparison is meaningless.

Why "redundancy rate instead rising" is not a bug. A natural reviewer question is: the active arm's redundancy rate 0.692 is instead higher than passive 0.599—does it show the active arm is in fact repeatedly collecting? Our explanation is: redundancy rate measures "the degree qualified clips mutually cover." When coverage is pushed to 215/226 (95%), the remaining selectable clips naturally can fill only very few new cells and overlap necessarily rises. This is the mathematical necessity of coverage saturation, not policy failure. The engineering decision truly to make is: take "coverage near saturation" as a stop signal, stop scaling volume when coverage reaches ~90% and turn to targeted re-collection of the 11 uncovered cells, rather than continuing to spend budget for redundancy.

Why the yield definition must be "blind." If QC could peek at defect ground-truth labels, then only "discard all clips injected with drift/static/duplicate" would be needed and yield would be artificially pushed near 1.0—but this cannot be done in real scenes because real scenes have no truth labels. Insisting on blind QC is to make this yield definition executable at real deployment. In the experimental setup we strictly limit QC to observable signals (trajectory length, jitter proxy, quantized signature) and confirm in the `quality_control` source that the defect truth `defect` field does not participate in the verdict.

8 Resource Gates and Applicability Boundaries

Currently reachable (CPU, completed in this paper): synthetic flywheel, blind QC, submodular greedy, three-arm control, coverage/density/redundancy metrics, 10 contract tests.

Gates needed to unlock real data (currently unreachable, not simulated):

- real egocentric collection network and hand tracking/SLAM;
- video-understanding large models ("MaxVLM-like," needing multiple LLM/vision API keys) for real QC;
- real-machine verification across the embodiment gap.

Planned factorial design (currently only single-cell pilot, full factorial not yet run). The full factorial the one-pager planned but this paper has not executed is: budget $\{12, 24, 48, 96\}$ clips, defect rate $\{0.1, 0.25, 0.4\}$, pool skew $\{\text{low, medium, high}\}$, state-map granularity $\{\text{coarse, fine}\}$. Current measurement takes only one cell (budget 48, defect rate 0.25, passive-skewed vs uniform-active). We expect: (i) the smaller the budget, the larger the active-vs-passive coverage advantage (because the skewed pool hits the wall earlier under a small budget); (ii) the higher the defect rate, the larger blind QC's impact on yield; (iii) the higher the skew, the more H3 holds. These are all **to-verify expectations**, not written as conclusions before the full factorial, marked `[RESULT NEEDED: full-factorial grid × budget/defect rate/skew/granularity]`.

Pre-registered analysis plan (ab-experiment-analysis gate):

- Randomization unit: independent seeds under the same world and defect rate; exposure unit = one complete run.
- Main metric: oracle coverage; guardrail metrics: yield, experience density, redundancy rate, late marginals.
- Minimum detectable effect (MDE): coverage absolute difference ≥ 0.05 ; significance $\alpha=0.05$, power 0.8; sample size estimated per baseline-variance scenario [RESULT NEEDED: multi-seed baseline variance $\rightarrow$ sample size].
- Stop/quality gates: SRM (three-arm pool sizes consistent), defect injection rate consistent, QC thresholds fixed without looking back; no tuning to remove negative results.
- Decision rule: only when the active arm's main metric is significantly better than passive and guardrails do not fall is H1 supported; the same-pool ablation significantly supports selection attribution.

9 Conclusion

This paper turned "coverage-aware egocentric data collection" from a slogan into a reproducible, falsifiable, oracle-bearing CPU benchmark. Under a fixed budget, blind QC + submodular greedy raised yield from 0.6667 to 0.8542, oracle coverage from 0.695 to 0.951, density from 7.80 to 11.38 (all seed=0 pilot point estimates); the same-pool ablation proved gains come jointly from the selection policy and pool de-biasing. We do not cross-validate with any vendor's "98% yield" but emphasize: yield must be blind, coverage must reach the state level, gains must have ablation attribution. The next step is on ≥ 30 seeds and small-scale real video annotation to upgrade directional assertions into CI-bearing confirmatory conclusions.

One-sentence conclusion: under a fixed collection budget, replacing passive volume scaling of "which experience segment to collect" with a submodular greedy over the coverage function and keeping the yield definition blind is an engineering means reproducible on CPU and directionally consistently improving yield, state coverage and experience density; its real value lies not in some pretty yield number but in turning data collection from "head-pat volume scaling" into a "measurable, stoppable, attributable" closed loop. We neither cross-validate with vendors' absolute yield nor call single-seed point estimates statistically significant—this restraint itself is the methodological stance this paper wants to convey: in a data collection team, first measuring yield, coverage, density with the same reproducible caliber, then discussing scale expansion, better avoids "volume without quality" than chasing absolute numbers from the start.

References

Note: the first two entries below are  verified items in this shared "Reference Master Library" (verified 2026-09-19, fields copied from the master); the rest are search directions to add, keeping placeholders, not forging sources.

Verified (2026-09-19)

1. Paul, M., Ganguli, S., Dziugaite, D. (2021). Deep Learning on a Data Diet: Finding Important Examples Early in Training (GraNd/EL2N). NeurIPS 2021. arXiv:2106.00278. (link: early gradient-norm scoring

pruning about half the data without hurting accuracy, the direct reference for this paper's data point selection.)

2. Krogh, A., Vedelsby, J. (1995). Neural Network Ensembles, Cross Validation, and Active Learning. NIPS 8 (NeurIPS 1995). (link: bias-variance-ambiguity decomposition and query-by-committee, the theoretical ancestor of this paper's "marginal coverage as information proxy.")

To verify (keeping [CITATION NEEDED] , not forging)

- Submodular coverage greedy and $1-1/e$ approximation:
submodular maximization coverage function greedy $1-1/e$ (candidate: Nemhauser/Wolsey/Fisher classic result).
- Robot learning trajectory subset/coreset: coreset selection robot learning trajectory subset.
- Egocentric/embodyed manipulation datasets: egocentric video dataset manipulation learning from human videos.
- Data cleaning and deduplication: data deduplication filtering pretraining dataset quality.
- Coverage-aware/active collection: active dataset acquisition coverage-aware robot data.

Appendix

Appendix A Reproduction Commands and Evidence Ledger

```
# repo: udos-engine tag v7.5.0 (commit 1a71270)
python scripts7/egodata_flywheel_demo.py      # generates reports7/egodata_flywheel_demo.json
pytest tests7/test_v740_flywheel.py -q        # 10 contract tests
```

Evidence ledger: - Main data: reports7/egodata_flywheel_demo.json (v7.4.0, cpu-proto), trace 7a8ca2ca4ce94bf0 .
- Source: udos7/egodata/{episodes,coverage,processing,collection,benchmark}.py. - Tests: tests7/test_v740_flywheel.py (10 items: determinism, QC rejecting three defect classes, state vs task coverage, density, same-pool active $\geq$ passive, yield and rejection counts, benchmark layering, sparsest-task guidance).

Reproduction consistency. tests7/test_v740_flywheel.py test_benchmark_active_covers_more_and_graded uses pool_size=50, budget=16, seed=11 to run the benchmark twice and asserts the two's state_cells are exactly equal (`= r2["..."]`), fixing the deterministic property "same seed same config results bitwise consistent" as a regression assertion. This means the seed=0 main numbers this paper reports, anyone rerunning under the same version code should reproduce bitwise, with no randomness-induced number drift. The trace record (trace_id=7a8ca2ca4ce94bf0, single 4889 ms, token_usage=0) also corroborates the benchmark calls no external LLM, purely local CPU computation.

1. Yield 0.8542 $\leftarrow$ JSON coverage_aware_collection.yield.
2. Oracle coverage 0.951 $\leftarrow$ JSON coverage_aware_collection.oracle_fraction (215/226 $\approx$ 0.9513).
3. Passive density 7.80 $\leftarrow$ JSON passive_collection.mean_density.
4. Ablation coverage 0.889 $\leftarrow$ JSON ablation_greedy_on_skewed_pool.oracle_fraction (201/226 $\approx$ 0.8894).
5. Duplicate rejections 8 $\leftarrow$ JSON passive_collection.reject_reasons.duplicate.

Appendix C Author's Intended Statements

- **Target outlets:** CoRL / RSS workshop, NeurIPS Datasets & Benchmarks, ICRA; journals IJRR / IEEE RA-L / T-PAMI (if containing real data). Quartiles/impact factors/deadlines all [to check], verified online before submission with the verification date marked.
- **Pre-registration plan:** per Section 8 main metrics/guardrails/MDE/stop rules, freezing and registering the plan before collecting ≥ 30 -seed data.
- **Data and code availability:** Apache-2.0; code tag v7.5.0 (commit 1a71270); report reports7/egodata_flywheel_demo.json.
- **AI use statement:** this paper's writing used AI assistance for draft generation and plotting scripts, all numbers re-sourced and checked by the author one by one from reports7 and source; external numbers keep unverified labels, no unverified p-values/CI/DOI/quartiles filled.

Appendix D Internal Review Record (five-dimension reviewer self-assessment, look-only)

Per doubao-academic-evaluator five-dimension self-assessment (1–5):

1. **Novelty:** 3/5. Submodular selection and blind QC are both mature, the novelty in "making egocentric yield/coverage into an oracle-bearing reproducible benchmark."
2. **Evidence strength:** 2/5 (current draft). All single-seed point estimates, no CI/p-values; this is the current largest hard flaw, honestly marked [RESULT NEEDED].
3. **Reproducibility:** 5/5. Fixed seed, CPU, reproduction commands, test index, evidence ledger complete.
4. **External validity:** 2/5. Purely synthetic proxy world, explicitly declared not equal to real video yield.
5. **Writing and honesty:** 4/5. Distinguishes verified/cpu-proto/unverified, negative results (redundancy rate instead rising, subset $\neq$ training gain) honestly presented.

Overall judgment: currently a **submittable method/benchmark draft skeleton, but evidence strength does not reach the bar**. Before submission one must complete ≥ 30 -seed paired statistics and at least a small-scale real video annotation pilot, otherwise it is recommended to position it as a benchmark/workshop short paper rather than a main-conference long paper.